

- 6.** Particles X (of mass 4 units) and Y (of mass 9 units) move directly towards each other, collide and then separate.

If Δv_x is the change in velocity of X and Δv_y is the change in velocity of Y , the magnitude of ratio $\frac{\Delta v_x}{\Delta v_y}$ is

A $9 / 4$

B $3 / 2$

C 1

D $2 / 3$